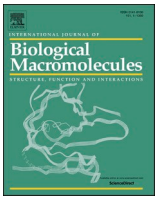
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The correlation between human seminal plasma sialoproteins and ejaculate parameters

Veronika Palenikova [a,b], Hana Pavlova [a], Daniel Kraus [c], Zuzana Kratka [d], Katerina Komrskova [a,e], Pavla Postlerova [a,f,*]

[a] Laboratory of Reproductive Biology, Institute of Biotechnology of the Czech Academy of Sciences, BIOCEV, Prumyslova 595, 252 50 Vestec, Czech Republic
[b] Department of Biochemistry, Faculty of Science, Charles University, Hlavova 8, 128 40 Prague 2, Czech Republic
[c] Laboratory of Gene Expression, Institute of Biotechnology of the Czech Academy of Sciences, BIOCEV, Prumyslova 595, 252 50 Vestec, Czech Republic
[d] Gennet Archa Palace, Prague, Czech Republic
[e] Department of Zoology, Faculty of Science, Charles University, BIOCEV, Vinicna 7, 128 44 Prague 2, Czech Republic
[f] Department of Veterinary Sciences, Faculty of Agrobiology, Food and Natural Resources, University of Life Sciences Prague, Kamycka 129, 165 00 Prague 6, Czech Republic

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ABSTRACT

Sialic acids are negatively charged carbohydrates that are components of saccharide chains covalently linked to macromolecules. Sialylated glycoproteins are important for most biological processes, including reproduction, where they are associated with spermatogenesis, sperm motility, immune responses, and fertilization. Changes in the glycoprotein profile or sialylation in glycoproteins are likely to affect the quality of ejaculate. The aim of this study was to determine differences in the degree of sialylation between normozoospermic ejaculates and ejaculates with a pathological spermiogram using two lectins, *Sambucus nigra* (SNA) and *Maackia amurensis* (MAL II/MAA) recognizing α-2,6 or α-2,3 linkage of Sia to galactosyl residues. Our results show a close relationship between seminal plasma (SP) sialoproteins and the presence of anti-sperm antibodies in the ejaculate, apoptotic spermatozoa, and ejaculate quality. Using mass spectrometry, we identified SP sialoproteins such as, semenogelins, glycodelin, prolactin-inducible protein, lactotransferrin, and clusterin that are associated with spermatozoa and contribute to the modulation of the immune response and sperm apoptosis. Our findings suggest a correlation between the degree of SP glycoprotein sialylation and the existence of possible pathological states of spermatozoa and reproductive organs. Glycoproteins sialylation represents a potential parameter reflecting the overall quality of ejaculate and could potentially be utilised in diagnostics.

1. Introduction

The number of couples with fertility disorders increases every year and currently impacts every sixth couple [1]. The quality of ejaculate is a critical parameter that reflects sexual competence and effects successful fertilization. The typical criteria used to determine the quality of the ejaculate are values recorded in the spermiogram, such as the volume of the ejaculate, the sperm count and concentration, or sperm motility, and morphological abnormalities. Often the boundary between normozoospermic (NS) and pathological samples may be ambiguous and therefore even changes in the NS group can indicate a developing pathological condition. Some laboratories diagnose the presence of anti-sperm antibodies (ASA) or apoptotic spermatozoa, which can also effect the final quality of the ejaculate [1]. In the present study, we evaluated sperm abnormalities based on the standardized spermiogram, ASA and sperm apoptosis, and also search for correlations between these values and Sia profile in SP proteins. The goal of this paper was to evaluate whether glycoprotein sialyation could become a novel marker to predict a pathological state of the ejaculate.

Sialic acids (Sia) are monosaccharide units and are common components of saccharide chains covalently bound to lipids or proteins. Glycoproteins are a heterogeneous group, which is an essential characteristic for biochemical and biological processes [2]. Sialic acids with their negative charge are important molecules involved in several

* Corresponding author at: Laboratory of Reproductive Biology, Institute of Biotechnology of the Czech Academy of Sciences, BIOCEV, Prumyslova 595, 252 50 Vestec, Czech Republic.
E-mail addresses: veronika.palenikova@ibt.cas.cz (V. Palenikova), pavla.postlerova@ibt.cas.cz (P. Postlerova).

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biological processes, including fertilization [3]. The role of sialylated glycoproteins in reproduction relates to their importance in spermatogenesis, sperm motility, immune response, and the fertilization process [4]. Experiments on mice show the sialome of sperm is developed from spermatogenesis, however, during sperm maturation in the epididymis the surface sialome harvest is modified [5]. Sialic acids are essential chemicals for sperm because they help sperm penetrate through the cervical mucus [6], suppress the immune system in the female genital tract through the interaction with Siglec receptors [7], and mask receptors for zona pellucida binding [8].

In this paper, we assume that the modification of sialoproteins in sperm and SP might lead to fertilization failure. Such a change can be caused, for example, by the presence of an infection or cancer. It was previously identified that the presence of bacterial infections such as *Escherichia coli* in the epididymis was associated with increased activity of endogenous sialidases, leading to a desialysation of sperm glycocalyx [9]. As an example, cancer is associated with changes in protein glycosylation including sialylation [10]. Specifically, testicular cancer is known to trigger the hyperglycosylation with a lower presence of sialyl T antigen of human chorionic gonadotropin (hCG), [11], and is linked to lower ejaculate quality [12]. Additionally, sialylated proteins of SP such as fibronectin, lactotransferrin, clusterin, and zinc-α2-glycoprotein are significantly down-expressed during prostate cancer [13,14].

Since Sia as glycan components are essential for sperm survival and successful fertilization [15], our main objective is to demonstrate that modification in SP protein sialylation relates pathological processes of the reproductive tract, such as the presence of ASA, sperm apoptosis, or impaired ejaculate quality. A recent study by Menevse et al. ([16]) found significantly lower Sia concentrations in NS samples than in oligozoospermic samples, supporting our hypothesis that Sia changes in SP proteins may indicate a pathological condition in the ejaculate. As a result, our study focused on sialoprotein detection in NS, asthenozoospermic (AS), and oligoasthenoteratozoospermic (OAT) ejaculate samples. Evidence that a modification of SP protein sialylation is related to a pathological state of the ejaculate may lead to improved diagnostic methods and improved therapeutic approaches, which may prove essential in the current era of increasing infertility disorders.

2. Material and methods

2.1. Sample preparation

Seminal plasma (SP) was obtained from the ejaculate of men after 3–4 days of sexual abstention, in the Immunological Laboratory, Gennet, Prague. Samples were approved by the Ethics Committee of the General University Hospital, Prague (671/17 S-IV) under informed consent and in accordance with the institute's Human Ethics Committee guidelines. Spermiograms, MAR (Mixed Anti-immunoglobulin Reaction) tests (IgA and IgG) and tests on apoptosis were performed with certified methods in the Immunological Laboratory at Gennet. Average values and standard deviations of clinical and laboratory parameters of patient samples can be reviewed in Supplementary Tables S1 and S4. Seminal plasma was centrifuged from spermatozoa in the Immunological Laboratory at Gennet at 300*g* for 10 min and was held in −80 °C until processing. After SP thawing on ice, the samples were centrifuged at 2000 ×*g* for 10 min at 4 °C. The supernatants were transferred into new tubes and samples were centrifuged at 10,000 ×*g* for 10 min at 4 °C. Protein concentration in SP samples was measured by Qubit Fluorometric Quantification (Invitrogen, Waltham, MA, USA). The supernatants were transferred into new tubes and used for further evaluation.

2.2. Lectins

Two lectins were used for the detection of Sia in SP: i) *Maackia Amurensis* Lectin II (MAL II/MAA) Biotinylated (B1265) for specific detection of α-2,3 bound Sia; ii) *Sambus Nigra* lectin (SNA), Biotinylated (B-1305), and *Sambus Nigra* lectin (SNA), Agarose Bound (AL-1303) for detection of Sia attached in linkage α-2,6 to terminal galactose in glycans. All lectins were received from Vector Laboratories (Burlingame, CA, USA) and the lectin concentration used for Enzyme-Linked Lectin Sorbent Assay (ELLSA) are summarized in Table 1 and for Western blot analysis was used concentration of lectins 1 μg/ml.

2.3. Enzyme-linked lectin sorbent assay (ELLSA)

Seminal plasma proteins at a concentration of 5 μg in 100 μl binding buffer (0.05 M $NaHCO_3$; pH 9.6) were applied to the well microtiter plate. The proteins were incubated at 4 °C overnight with constant shaking. After incubation, the bound proteins in wells were washed for 3 × 10 min by phosphate-buffered saline (PBS) with 0.05 % Tween (T-PBS) at room temperature (RT) with constant shaking. Wells were blocked by 1 % gelatine from cold-water fish skin (9000-70-8; Sigma-Aldrich®, St-Louis, MO, USA) for 1 h at RT with constant shaking. After the blocking, the wells were washed 3 × 10 min with T-PBS at RT and 100 μl of biotinylated lectin were added to the wells diluted in HEPES buffer (10 mM HEPES; 0.1 mM $CaCl_2$; 0.15 M NaCl; pH 7.5). Used concentrations of lectins are listed in Table 1. The lectins were incubated with the proteins bound in the wells for 1 h at RT with constant agitation. The wells were washed for 3 × 10 min with T-PBS at RT and 100 μl Avidin labelled with peroxidase (Sigma-Aldrich®; A-3151) diluted 1:2000 in PBS (with a final concentration of 0.5 μg/ml) was added into the wells. Incubation was carried out 1 h at RT with constant shaking. The wells were washed for 3 × 10 min in T-PBS and 100 μl with 3,3′,5,5′-Tetramethylbenzidine Liquid Substrate System for ELISA (Sigma-Aldrich®; T0440) was added to the wells. Incubation was performed for 10 min at RT in the dark. After 10 min, the reaction was stopped with 1 N HCl and the intensity of the colorimetric staining was measured at 450 nm by microplate reader (Tecan, Grödig, Austria) within 30 min after the addition of the stop solution.

2.4. Lectin affinity chromatography

Agarose-bound SNA lectin was loaded on Pierce™ Spin Columns (Thermo Scientific™, Waltham, MA, USA) and washed 3 × with HEPES buffer. After washing, a mix of SP from five samples with protease inhibitors was added to the column followed by incubation for 3 h at RT with constant shaking on rotator. Next, the contents of the columns were centrifuged at 1000 ×*g* for 2 min at RT. The supernatant was kept frozen, and the column was washed with HEPES buffer three times. Subsequently, the glycoproteins bound on SNA lectin were eluted with Glycoprotein Eluting Solution for Sialic Acid Binding Lectins (ES-7100; Vector Laboratories), and centrifuged at 1000 ×*g* for 2 min at RT. The supernatants containing released glycoproteins were loaded on Pierce™ Protein Concentrators PES (3 K MWCO; Thermo Scientific™), and the elution buffer was exchanged for PBS buffer. The glycoproteins in PBS were precipitated by frozen acetone at 1 h for −20 °C followed by centrifugation at 7000 ×*g* for 15 min. The glycoprotein pellet was resuspended in a non-reducing Laemmli sample buffer [20 % glycerol, 4 % Sodium Dodecyl Sulphate (SDS), 0.125 M Tris-HCl, pH 6.8, 0.005 % bromophenol blue] and subjected to the SDS-PAGE.

2.5. SDS electrophoresis and Western blot analysis

The purified SP (see chapter Sample Preparation) was diluted 1:1 in

Table 1
Used lectins with their dilutions for ELLSA (Enzyme-linked lectin sorbent assay) method.

Sialic acids binding by biotinylated lectins	Used concentration for ELLSA		
Maackia Amurensis Lectin II (MAL II/MAA)	6 μg/ml	3 μg/ml	1 μg/ml
Sambus Nigra Lectin (SNA)	1 μg/ml	0.5 μg/ml	0.25 μg/ml

non-reducing sample solutions with protease inhibitors Complete Mini (11836153001; Sigma-Aldrich®). The amount of 10 μg of SP proteins and purified glycoproteins were loaded and separated by 10 % SDS-PAGE. The gels Mini-PROTEAN TGX Gels 4–20 % (456–1096; Bio-Rad, Hercules, CA, USA;) used for samples appointed to the identification by mass spectrometry (MS) were washed in dH_2O and stained in Coomassie Brilliant Blue solution (0.25 g Coomassie Brilliant Blue; 35 ml acetic acid; 250 ml ethanol; 215 ml dH_2O) at RT overnight. The next day gels were incubated in the destaining solution (350 ml ethanol; 100 ml acetic acid; 550 ml dH_2O) until the background was decolorized. Next, the protein bands were cut, placed into 1 % acetic acid, and subjected to MS analysis. The molecular weight of the proteins was assigned by Precision plus Protein Dual Colour Standards (Bio-Rad).

For Western blot (WB) analysis with Sia detection, the separated proteins in gels were transferred on the Hybridization Nitrocellulose membrane (Millipore, Darmstadt, Germany) and blocked in 1 % gelatine from cold water fish skin (9000-70-8; Sigma-Aldrich®) for 1 h at RT. Afterwards, the membrane was incubated at 4 °C overnight with biotin-labelled lectins (SNA and MAL II/MAA) diluted in HEPES buffer to the final concentration of 1 μg/ml concentration. Then, the membrane was washed in T-PBS and incubated with Avidin conjugated with Horseradish Peroxidase (A-3151; Sigma-Aldrich®) diluted 1:1000 (final concentration 1 μg/ml) in PBS for 1 h at RT, and consequently washed in T-PBS. As a negative control, a membrane only was incubated with Avidin-Proxidase. The reaction was visualized with SuperSignal West Pico Chemiluminescent Substrate (Thermo Scientific™, Rockford, IL, USA) and the signal was documented by Azure Imaging Systems (Azure Biosystems Inc., Dublin, CA, USA).

2.6. Data analysis

The densitometry analysis was measured by the Image J system (Java-based image processing program, LOCI, University of Wisconsin, WI, USA). Statistical analysis was performed by GraphPad Prism 9.3.1. (GraphPad Software, Inc., La Jolla, CA, USA). Original data from the ELLSA and densitometry analysis measurement are available as external material (https://zenodo.org/uploads/10801949) including the results from mass spectrometry. Descriptive statistics, which are part of the supplement, were prepared for the values obtained from the measurements as well as from laboratory parameters of individual patients. Since we obtained ordinal data, with a lower number of samples and we did not assume a Gaussian distribution of the data, we chose non-parametric statistical tests such as the Kruskal–Wallis test for comparing data between groups, and Spearman r test for correlation analysis, which are more suitable for the given data compared to the parametric tests. Mass spectrometry data were filtered based on the identified unique peptides together with their percentage of sequence in detected proteins. Proteins with zero unique peptides and low amino acid sequence coverage were not included in the analysis. We used the Matplotlib-Venn 0.11.6 package in Python 3.8 for visualization of unique and matched proteins between categories of samples. Gene ontology analysis was performed to assign molecular functions and biological processes for all identified proteins using Python 3.8 in the package Bioservices 1.8.0. We filtered the proteins based on specific functions, such as immune response, apoptosis, and association with spermatozoa. After the filtering in the Pandas 1.3.3 package in Python 3.8, we analysed the differences of individual sample groups. Proteins with specific functions were visualized using the package Matplotlib 3.4.3 in Python 3.8.

3. Results

We investigated the level of sialylated proteins in human SP and the role of protein sialylation in male fertility. We detected Sia in SP proteins linked by i) α-2,3 (recognized by MAL II/MAA), and ii) α-2,6 (recognized by SNA lectin) bonds. We used Enzyme-linked lectin sorbent assay (ELLSA), WB analysis, and affinity chromatography followed by MS identification of sialoproteins (Fig. 1). In total, our study included 43 patients with, NS, AS, and OAT analysed in ELLSA ($n = 29$) and WB ($n = 28$). For total protein sialylation detected by ELLSA method, the average age of the patients was 37 ± 5.457 years (NS: 36 ± 5.685, AS: 39 ± 5.862, OAT: 35 ± 4.664). Descriptive statistics of clinical parameters for patients used in the ELLSA method is part of Supplementary Table S1 and the values measured in the ELLSA method is part of Supplementary Table S2. The average age of patients used in the WB analysis was 38 ± 6.34 years (NS: 40 ± 8.573, AS: 38 ± 6.553, OAT: 36 ± 2.500), while descriptive statistics for laboratory parameters are part of Supplementary Table S4 and descriptive statistics of measured values are part of Supplementary Tables S5 and S6. The p and r value from correlation analysis for the ELLSA method is a part of the Supplementary Table S4 and for WB analysis are part of the Tables S7 and S8.

3.1. Changes in total Sia bound to SP glycoproteins

The total level of bound Sia in SP glycoproteins of normal and pathological spermiograms (summarized in Supplementary Table S1) was measured in ELLSA using two lectins. Average values and standard deviations of optical density (OD) measured for Sia are shown in Supplementary Table S2. We compared the protein sialylation between NS, AS, and OAT samples. AS samples showed a higher rate of sialylation in SP glycoproteins than NS and OAT samples, but the differences between groups were not significant (Fig. 2). We observed a similar intensity between MAA and SNA lectin detection; however, MAA binding showed greater variances between samples in individual spermiogram groups (Fig. 2a).

We investigated whether the values of protein sialylation obtained from ELLSA correlated with the laboratory parameters of ejaculate summarized in Supplementary Table S1. In the case of SNA lectin, we did not detect any relationship between ejaculate values. On the other hand, MAA lectin showed significant correlations with spermiogram values, specifically with ASA, late apoptosis, and morphological abnormalities (Supplementary Table S3). The correlation plots, with the linear dependence indicating the strength of the correlation for NS and OAT samples (Fig. 3). Intensity of protein sialylation showed a positive correlation with the percentage of sperm in late apoptotic stage for NS group (Fig. 3a), and with positive IgG ASA in ejaculate of OAT samples (Fig. 3b). AS samples did not show the required significant dependence in total protein sialylation in the studied spermiogram category (Supplementary Table S3).

3.2. Changes in sialylation of individual SP glycoproteins

After using the ELLSA method to understand the total sialylation of SP proteins, we focused on the variability in sialylation between individual proteins. Accordingly, we performed WB analysis with lectin detection as an additional method to give us an insight into the sialylation specificities of proteins characterized by different molecular sizes. We focused on more specific sialoprotein bands obtained by WB analysis with MAA and SNA lectins which appeared uniformly in all samples. The lectin signals of individual protein bands were evaluated by densitometry analysis. The densitometry values were related to the signal of NS sample with the following spermiogram parameters: non detectable ASA; 71.6 % viable sperm, 64 mil sperm/ml; minimum 50 % motile sperm, 15 % morphologically normal sperm, and 4 ml of ejaculate. The average values of the spermiograms are listed in Supplementary Table S4. The average values and standard deviations of relative optical density (ROD) measured for Sia detected by MAA and SNA lectins are shown in Supplementary Tables S5 and S6. We were also interested in which sialoproteins are were present in SP. To this end, we isolated sialylated SP glycoproteins from mixture samples of each group of NS, AS, and OAT using SNA lectin affinity chromatography. The purified SP sialoproteins were electrophoretically separated and selected protein bands based on WB analysis were subjected to MS analysis. The list of all

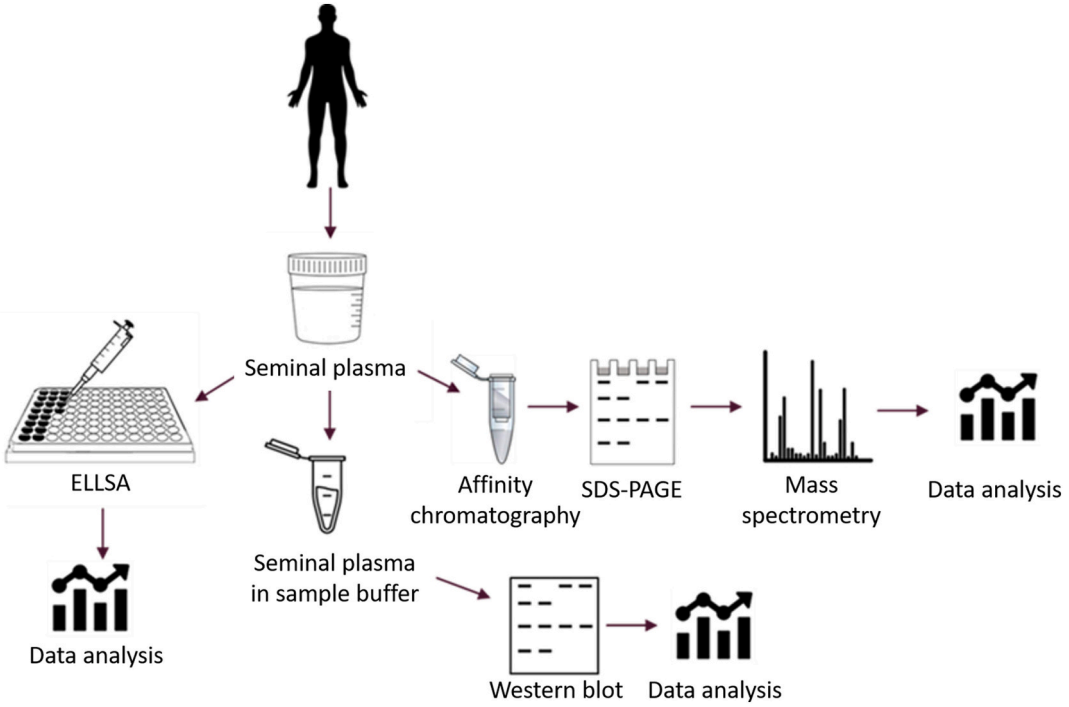


Fig. 1. Scheme describing the design of experimental study and use of selected methods.

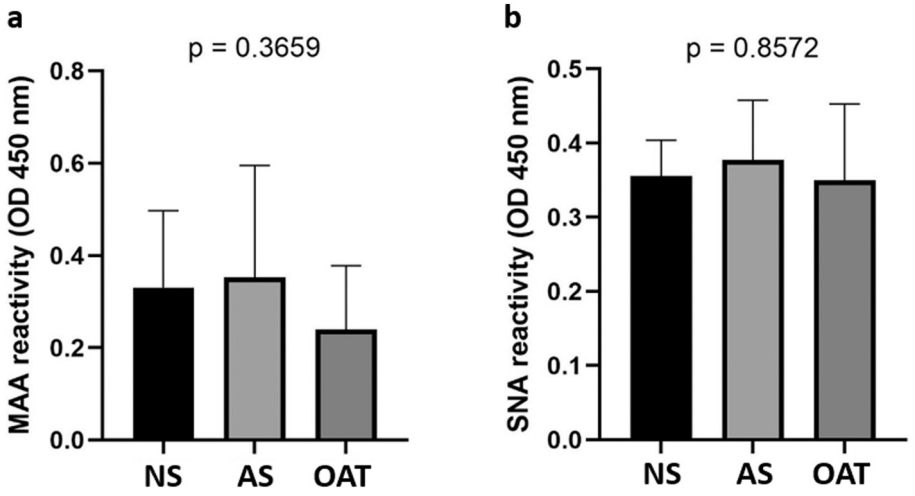


Fig. 2. Evaluation of SP protein sialylation in ELLSA tests. (a) Detection of α-2,3-bound Sia in SP proteins labelled with MAA lectin, and (b) detection of α-2,6-bound Sia in SP proteins with SNA lectin were not significant ($p > 0.05$) between NS samples and pathological (AS, OAT) samples. For statistical analysis, the Kruskal–Wallis test was performed.

identified proteins from the protein bands were stored in a database (as external material: https://zenodo.org/records/10356133) and are available for further analysis. From the identified sialoproteins, we determined proteins involved in the immune response, the apoptotic processes, or that were associated with spermatozoa.

MAA lectin labelled several protein bands, five of them (122, 96, 55, 37, and 25 kDa) were the most frequently sialoproteins in SP samples (Fig. 4a). Graphs (Fig. 4b–f) represent densitometry analysis for individual protein bands detected in SP. Statistical analysis did not show differences in the sialylation of SP glycoproteins between samples with different spermiograms ($p > 0.05$) (Fig. 4b–f).

Compared to MAA, the SNA lectin detected more proteins and with a stronger intensity. We obtained several glycoprotein bands with different molecular masses, from which we selected five bands (142, 65, 47, 40, and 23 kDa) for further analysis (Fig. 5a). Differences in sialylation of SP glycoproteins between groups were not significant ($p > 0.05$), although different sialylation was detected between the individual groups (Fig. 5b–f). Venn diagrams (Fig. 5b–f) showed the proportion of sialylated glycoproteins in the protein bands between the NS, AS, and OAT groups. Venn diagrams represent only the number of isolated sialylated glycoproteins in SP. The data inside the closed curves determine the number of equal proteins found within groups. The SNA signal at the 142 kDa band showed the highest intensity in OAT samples and the lowest in AS samples. However, the Venn diagram shows that the number of sialylated proteins is the same for both AS and OAT groups. The least sialoproteins were detected in NS samples (Fig. 5b). In the 65 kDa band, the SNA binding intensity was the lowest in the NS sample group; however, the number of identified sialoproteins was the highest.

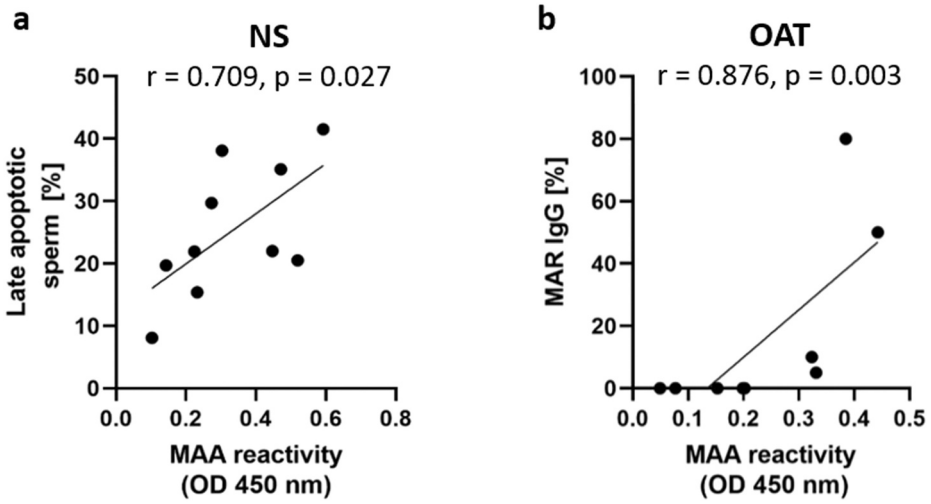


Fig. 3. Relationship between laboratory parameters of ejaculates and total protein sialylation detected by MAA lectin in SP. Scatterplots show values from significant results summarized in Supplementary Table S3. (a, b) Our results indicate that sialylation of SP proteins increased with the sperm apoptosis and the presence of ASA in the ejaculate.

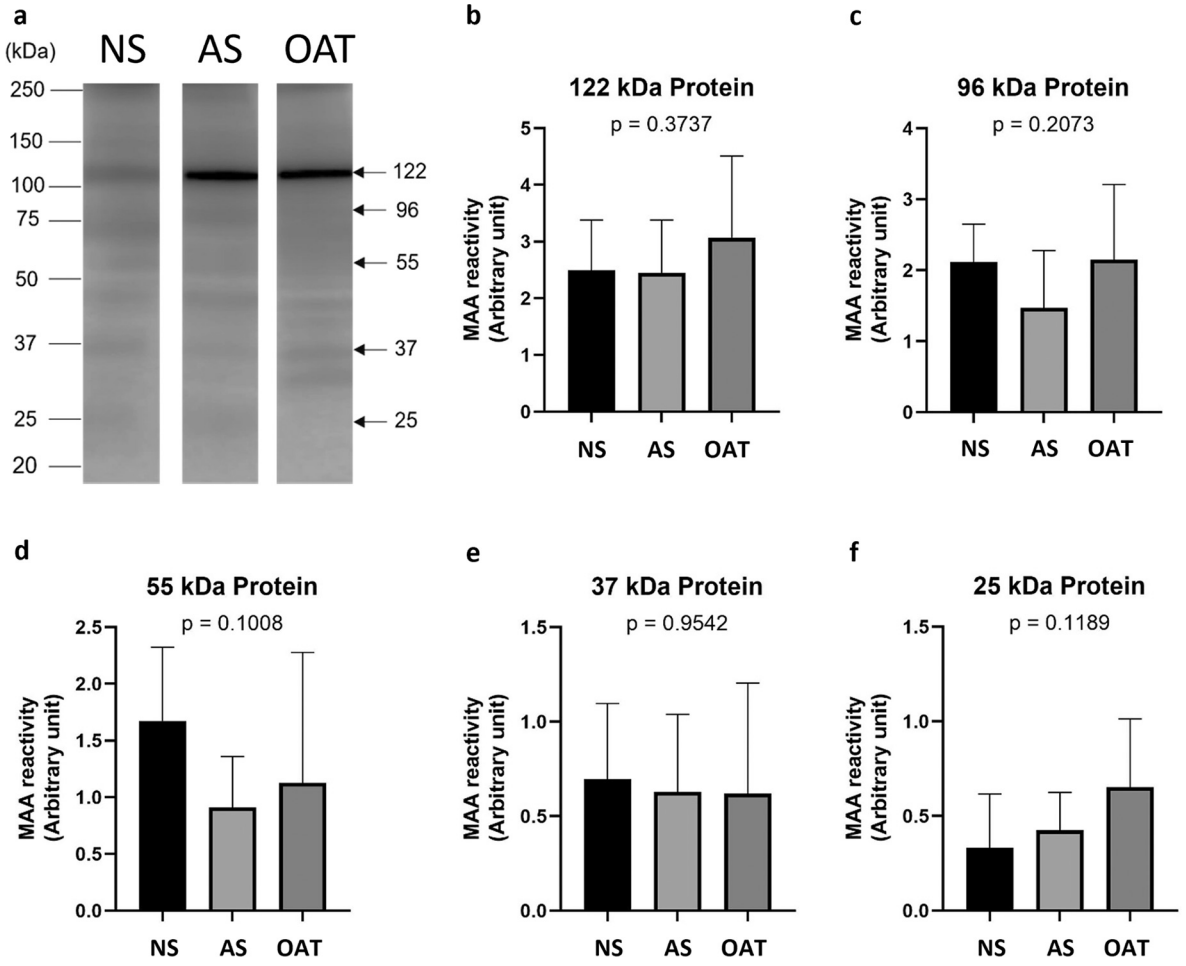


Fig. 4. Detection and evaluation of sialylated SP proteins detected by MAA lectin. (a) WB analysis detected several SP glycoproteins, wherein the proteins used for further analysis are indicated by arrows. (b–f) The comparison of the signal strength of the individual proteins between the spermiogram groups was not significant ($p > 0.05$), although different sialylation was detected between the individual groups. For statistical analysis, the Kruskal–Wallis test was performed. NS – normozoospermia (n = 10), AS – asthenozoospermia (n = 10), OAT – oligoasthenoteratozoospermia (n = 9).

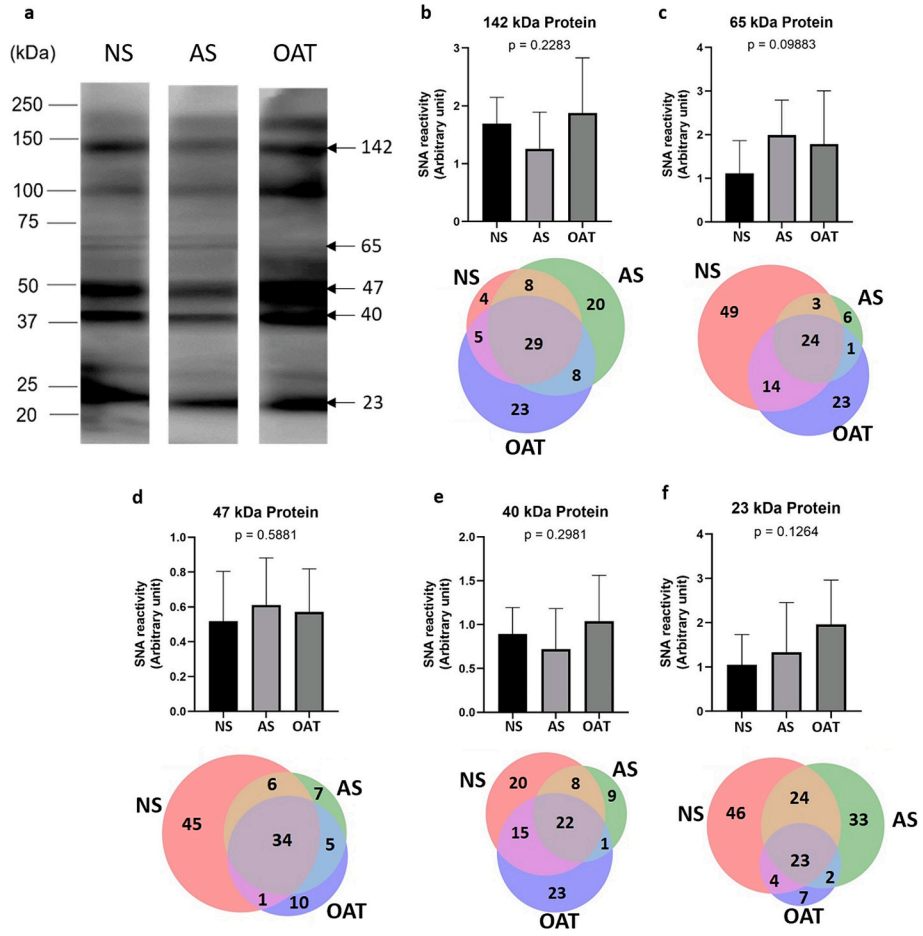


Fig. 5. Detection and evaluation of sialylated SP proteins detected by SNA lectin and purification of sialylated glycoproteins by agarose-bound SNA lectin. (a) WB analysis detected several glycoproteins of SP, wherein the proteins used for further analysis are indicated by arrows. (b–i) The comparison of the signal strength of the individual proteins between the spermiogram groups was not significant ($p > 0.05$). For statistical analysis, the Kruskal-Wallis test was performed. The Venn diagrams represent the number of identified sialoproteins for every protein band. NOR – normozoospermia (n = 10), AS – asthenozoospermia (n = 10), OAT – oligoasthenoteratozoospermia (n = 9).

Conversely, AS samples had the fewest number of sialylated proteins, but the SNA signal was the highest (Fig. 5c). For the 47 kDa band, approximately the same intensity was measured in all SP groups, but according to the Venn diagram, the most sialylated proteins were identified in NS samples followed by AS and OAT (Fig. 5d). The 40 kDa protein band had the highest intensity of SNA signal in AS samples, which agreed with the number of identified proteins. The highest intensity was obtained in OAT samples, however, this is not obviously different from NS samples, and the number of detected proteins is similar (Fig. 5e). The SNA binding intensity of the last studied protein band (23 kDa) was lowest in NS samples and highest in OAT samples. However, the number of identified sialoproteins increases with decreasing ejaculate abnormality (Fig. 5f).

3.3. Correlation between sialylation of SP proteins and ejaculate quality

Since the correlation analysis of the values from the ELLSA method showed significant dependencies between sialylation detected by MAA and ejaculate abnormalities (Fig. 3), we were interested in the correlation analysis of densitometry data from WB and data from ejaculate parameters. We were particularly interested in those parameters, which showed a positive relationship between sialylation of SP proteins detected by MAA or SNA lectin and ASA presence in ejaculate (Fig. 6, Supplementary Tables S7, S8). Fig. 6a represents the *r* value from correlation tests showing significant results highlighted in boxes with dashed borders and asterisks. Only Sia detected in the NS group has a stronger positive correlation with increasing ASA in ejaculate. For MAA detection, the 55 kDa protein band was found with a significant correlation with IgA ASA. SNA lectin detected 47, 40 and 23 kDa bands which were in positive correlation with IgA or IgG ASA. Correlation graphs are part of the supplementary material (Supplementary Fig. S1).

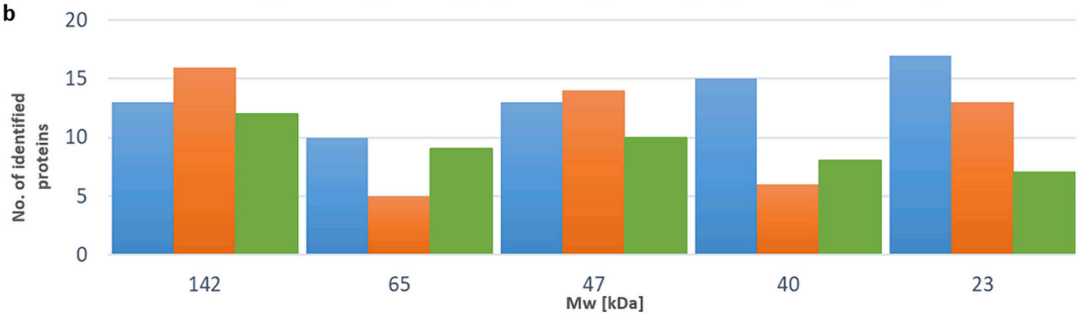
Using affinity chromatography and MS, we identified sialylated SP proteins. Using gene ontology, we managed to select proteins that play a role in the immune response (Fig. 6b). The graph represents the number of identified proteins in individual spermiogram groups and the table below the graph shows the list of selected proteins for each group. The 47 kDa sialylated band in the NS group with IgG and IgA ASA gave a positive correlation and contained semenogelin 1 (SEMG 1) in contrast to pathological samples without correlation. Moreover, the correlation value in the relationship between Sia and IgG ASA was higher than for the amount of diagnosed IgA. On the other hand, only in pathological samples protein S100-A9 (S10A9) and CD177 antigen were detected. The 40 kDa and 23 kDa bands, which were in positive correlation with IgG and IgA ASA in the NS group, respectively, had a higher presence of sialylated glycoproteins playing a role in immune response than pathological groups of samples. In both mentioned sialylated bands in NS samples, clusterin (CLU) was identified. Additionally, only in the NS group did the 40 kDa band contain alpha-1-acid glycoprotein 1 (A1AG1), arginase-1 (ARGI1), ribonuclease T2 (RNT2), matrin-3 (MATR3), and the 23 kDa band Protein S100-A8 (S10A8), neutrophil gelatinase-associated lipocalin (NGAL), OTU domain-containing protein 7B (OTU7B), neutrophil elastase (ELNE) and protein THEMIS (THMS1).

a

Detection	α-2,3 linkage of sialic acid (MAA)						α-2,6 and α-2,3 linkage of sialic acid (SNA)					
Method	ELLSA	Western Blot Analysis					ELLSA	Western Blot Analysis				
Protein Sia / MAR Test	Total	122 kDa	96 kDa	55 kDa	37 kDa	25 kDa	Total	142 kDa	65 kDa	47 kDa	40 kDa	23 kDa
IgA	0.226	0.115	0.488	* 0.692	0.257	0.044	0.000	-0.293	0.257	* 0.727	0.319	* 0.745
IgG	0.494	0.145	0.043	0.248	0.351	-0.017	0.532	-0.274	0.368	*** 0.949	* 0.787	0.650
IgA	0.086	0.006	-0.205	-0.068	0.031	-0.565	-0.419	0.131	-0.019	-0.006	-0.093	0.031
IgG	-0.138	-0.185	-0.165	-0.247	-0.185	-0.487	-0.046	-0.069	0.233	0.096	-0.165	0.123
IgA	0.383	0.487	0.530	0.325	0.482	0.162	0.420	-0.316	-0.137	-0.171	0.026	0.086
IgG	* 0.876	0.317	0.406	0.099	0.187	0.158	0.274	-0.386	-0.099	-0.158	0.020	0.307



b



NS	AS	OAT	NS	AS	OAT	NS	AS	OAT	NS	AS	OAT	NS	AS	OAT
TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL
DCD	DCD	DCD	DCD	DCD	DCD	DCD	DCD	DCD	DCD	DCD	DCD	DCD	DCD	DCD
K1C16	K1C16	K1C16	K1C16	K1C16	K1C16	K1C16	K1C16	K1C16	K1C16	-	K1C16	K1C16	K1C16	K1C16
K2C6A	K2C6A	K2C6A	K2C6A	K2C6A	K2C6A	K2C6A	K2C6A	K2C6A	K2C6A	K2C6A	-	K2C6A	K2C6A	K2C6A
PIP	PIP	PIP	PIP	-	PIP	PIP	PIP	PIP	PIP	PIP	PIP	PIP	PIP	PIP
S10A9	S10A9	S10A9	S10A9	-	S10A9	S10A9	S10A9	-	S10A9	-	S10A9	S10A9	S10A9	S10A9
CLUS	CLUS	CLUS	CLUS	CLUS	CLUS	CLUS	CLUS	-	CLUS	-	-	CLUS	-	-
-	G3P	G3P	G3P	-	-	G3P	G3P	G3P	G3P	-	G3P	-	-	-
S10A7	-	S10A7	S10A7	-	-	-	S10A7	S10A7	-	-	-	S10A7	S10A7	-
-	-	S10A8	-	-	-	S10A8	S10A8	S10A8	S10A8	-	S10A8	S10A8	-	-
-	ARGI1	ARGI1	ARGI1	-	-	-	-	-	ARGI1	-	-	-	ARGI1	-
-	-	-	-	-	-	SEMG1	-	-	SEMG1	SEMG1	SEMG1	SEMG1	SEMG1	SEMG1
-	-	-	-	-	ECM1	ECM1	ECM1	-	-	-	-	-	-	-
-	-	-	-	-	HEMO	-	-	-	HEMO	HEMO	-	-	-	-
-	-	-	-	-	-	A1AG1	A1AG1	A1AG1	A1AG1	-	-	-	-	-
-	-	-	-	-	-	-	CD177	CD177	-	-	-	CD177	CD177	-
IGHG2	IGHG2	-	-	-	-	-	-	-	-	-	-	-	-	-
IGHG1	-	IGHG1	-	-	-	-	-	-	-	-	-	-	-	-
-	IGHA1	-	-	-	-	-	-	-	-	-	-	-	-	-
-	IGHA2	-	-	-	-	-	-	-	-	-	-	-	-	-
-	IGLC2	-	-	-	-	-	-	-	-	-	-	-	-	-
-	IGLC3	-	-	-	-	-	-	-	-	-	-	-	-	-
-	IGHG3	-	-	-	-	-	-	-	-	-	-	-	-	-
-	IGHG4	-	-	-	-	-	-	-	-	-	-	-	-	-
IGKC	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4A	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4B	-	-	-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	A1AG2	A1AG2	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-	RNT2	-	-	-	-	-
-	-	-	-	-	-	-	-	-	MATR3	-	-	-	-	-
-	-	-	-	-	-	-	-	-	-	-	-	SAMP	SAMP	-
-	-	-	-	-	-	-	-	-	-	-	-	NGAL	-	-
-	-	-	-	-	-	-	-	-	-	-	-	CRIS3	CRIS3	-
-	-	-	-	-	-	-	-	-	-	-	-	OTU7B	-	-
-	-	-	-	-	-	-	-	-	-	-	-	ELNE	-	-
-	-	-	-	-	-	-	-	-	-	-	-	THMS1	-	-
-	-	-	-	-	-	-	-	-	-	-	-	-	GGT1	-

(caption on next page)

Fig. 6. Correlation between sialylation of SP proteins and ASA detected in ejaculate, and identification of proteins associated with immune response. (a) The r value from correlation test is determined for each group separately (blue – NS category, orange – AS category, green – OAT category). Significant results are highlighted with a black box and the strength of significance is shown with an asterisk (*$p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). (b) The bar plots represent number (No) of proteins in normal (NS) and pathological (AS, OAT) ejaculate samples of proteins bands in different molecular weights. TRFL – Lactotransferrin, DCD – Dermcidin K1C16 – Keratin, type I cytoskeletal 16, K2C6A – Keratin, type II cytoskeletal 6A, PIP – Prolactin-inducible protein, S10A9 – Protein S100-A9, CLUS – Clusterin, G3P – Glyceraldehyde-3-phosphate dehydrogenase, S10A7 – Protein S100-A7, S10A8 – Protein S100-A8, ARGI1 – Arginase-1, SEMG1 – Semenogelin-1, ECM1 – Extracellular matrix protein 1, HEMO – Hemopexin, A1AG1 – Alpha-1-acid glycoprotein 1, CD177 – CD177 antigen, IGHG2 – Immunoglobulin heavy constant gamma 2, IGHG1 – Immunoglobulin heavy constant gamma 1, IGHA1 – Immunoglobulin heavy constant alpha 1, IGHA2 – Immunoglobulin heavy constant alpha 2, IGLC2 – Immunoglobulin lambda constant 2, IGLC3 – Immunoglobulin lambda constant 3, IGHG3 – Immunoglobulin heavy constant gamma 3, IGHG4 – Immunoglobulin heavy constant gamma 4, IGKC – Immunoglobulin kappa constant, CO4A – Complement C4-A, CO4B – Complement C4-B, A1AG2 – Alpha-1-acid glycoprotein 2, RNT2-Ribonuclease T2, MATR3 – Matrin-3, SAMP – Serum amyloid P-component, NGAL – Neutrophil gelatinase-associated lipocalin, CRIS3 – Cysteine-rich secretory protein 3, OTU7B – OTU domain-containing protein 7B, ELNE – Neutrophil elastase, THMS1 – Protein THEMIS, GGT1 – Glutathione hydrolase 1 proenzyme.

Whether sialylation of these specific individual proteins or a group of sialylated proteins is responsible for the correlation cannot be determined.

Apoptosis of spermatozoa results in a decrease of fertilization ability. Our results showed that the change in SP protein sialylation in the AS group is associated with early apoptosis, especially with the 55 kDa protein band detected by MAA lectin and the 142, 47, and 40 kDa sialoproteins recognized by SNA lectin (Fig. 7a) and significant results are marked by a black dashed box with an asterisk indicating strength of significance. It should be noted that the correlation among individual samples was strongly positive (Supplementary Fig. S2). Using MS, we identified RAF proto-oncogene serine/threonine-protein kinase (RAF1), peroxiredoxin-2 (PRDX2), and cathepsin D (CATD) proteins in 142 kDa sialylated band, which were present only in the AS group of SP samples. Unlike NS and OAT, the 47 kDa protein band in AS samples contained polyubiquitin-B (UBB) and beta-2-glycoprotein 1 (APOH) between sialylated proteins in SP. Interestingly, the 40 kDa band had fewer identified proteins which were associated with apoptotic cells in SP, but metalloproteinase inhibitor 1 (TIMP1) was only detected in this molecular weight of AS samples (Fig. 7b).

The presence of spermatozoa in late apoptosis found in ejaculates was in strong correlation with the sialylation of SP protein in the NS group for the 23 kDa protein band detected by SNA (Supplementary Fig. S2). Compared to the pathological sperm groups, the NS group contained proteins in molecular size such as CLUS, S10A8, CATD, protocadherin gamma-C5 (PCDGM), protocadherin gamma-C3 (PCDGK), NGAL and phosphatidylethanolamine-binding protein 4 (PEBP4) (Fig. 7b). It remains to be determined whether the overall sialylation of SP proteins or sialylation of specific proteins is responsible for the observed correlations.

While we aimed to emphasize the significance of identified ASA and sperm apoptosis with regards to the sialylation of SP proteins, we questioned whether alterations in SP protein sialylation also correlated with the ejaculate basic parameters. One of the standard parameters is the evaluation of the ejaculate volume, which positively correlated with sialylation of SP proteins. This relationship was detected only with SNA lectin for the 47 and 40 kDa bands in NS group samples, where the higher sialylation correlated with higher volumes of ejaculate (Fig. 8a, Supplementary Fig. S3). MS identified the most proteins, including SEMG2, which were associated with spermatozoa in the NS group for the 47 and 40 kDa protein bands. Moreover, for the NS group within the 47 kDa band there was also detected specific SEMG1, and the 40 kDa band contained epididymal sperm-binding protein 1 (ESPB1) (Fig. 8b).

Other parameters used to diagnose ejaculate quality are sperm count and concentration, morphological abnormalities, and viability. Interestingly, we were able to find correlation between SP protein sialylation and lower sperm count and concentration only in OAT groups, particularly for increased Sia bound by α2-3 linkage for proteins in molecular weights of 122, 96, 37, and 25 kDa (Fig. 8a, Supplementary Fig. S4). MAA and SNA lectins detected protein bands in molecular weights of 55 and 23 kDa, respectively, in the NS group, which were in negative correlation between sialylation of SP proteins and morphological abnormalities of spermatozoa (Fig. 8a). MS showed plasma serine protease inhibitor (IPSP), cysteine-rich secretory protein 1 (CRIS1), mitochondrial inner membrane protease subunit 2 (IMP2L) in NS group comparing to pathological groups (Fig. 8b, as a list of proteins under the graph). Additionally, we found a correlation between the sialylation of the 55 kDa protein band detected by MAA lectin and sperm viability, indicating that increased sperm viability is associated with a reduced sialylation of proteins in SP (Fig. 8a, Supplementary Fig. S5).

From the identified sialylated proteins of SP, we selected those based on their functional properties classified by gene ontology and role in the organism. Many of these proteins were identified as glycoproteins, from which some of them contained Sia. These proteins play a role in the immune response (IR), apoptosis (Apo) or are associated with sperm function and successful fertilization (Rep). Table 2 represents a set of selected proteins, lactotrasferrin, prolactin-inducible protein (PIP), clusterin, glycderaldehyde-3-phosphate dehydrogenase, peroxiredoxin-2, semenogelin 1, semenogelin 2, CD177 antigen, glycodelin, keratin type I cytoskeletal 9, acrosin and epididymal sperm-binding protein 1. Many of these proteins do not yet have an assigned function in the reproductive process in the database, but some research articles have already demonstrated their potential role in fertilization (Table 2). In addition, there are possible mutual interactions between these identified sialylated proteins (Supplementary Fig. S6) and following proteins, which are most involved in the predicted interactions SEMG1, CLUS (*CLU*) and TRFL (*LTF*). On the contrary, there are no predicted interaction between PAEP, K1C9 (*KRT9*), ACRO (*ACR*) and ELSPB1 (*ELSPBP1*) proteins.

4. Discussion

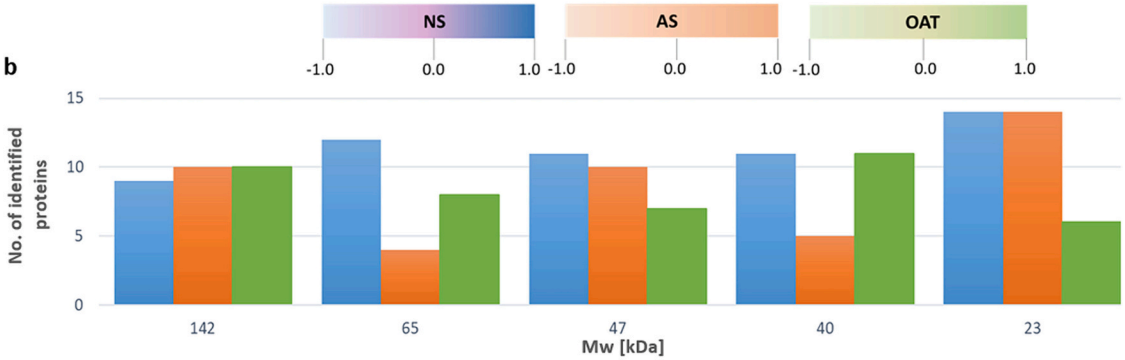
Proteins contained in SP are essential for spermatozoa which reflect the quality of ejaculate and sperm fitness [17]. Many SP proteins are glycoproteins with Sia in their oligosaccharide chains [18], which can be detected by lectins. In this research, less frequent protein sialylation with low signal intensity was detected by MAL II/MAA lectin (recognizing α-2,3 linkage) rather than SNA lectin (recognizing α-2,6 linkage). This is consistent with findings by Kałuża et al. [13] in the WB study and Kratz et al. [19] using the ELLSA method in NS, with variable outcome from sperm pathological samples.

Differences between NS samples and pathological samples were previously found [13,18,19] and we also detected the presence of sialylated proteins in SP by both ELLSA and WB analysis however, no significant differences in the total amount of sialyation between NS samples and pathological ejaculate samples were found. As expected, the intensity of the detected Sia signals differed between individual patients, especially in patients with a pathological spermiogram, despite careful protocol standardization for protein concentration and uniform conditions of experiments and analysis. This finding was supported by MS data, confirming that the number of sialylated proteins in SP varies between individual groups, and probably affects the final detected signal for Sia. The 23 kDa band had the highest signal intensity for OAT samples, but the number of identified proteins was 36, while for NS group the same band showed the lowest intensity with 97 identified proteins. Therefore, focusing on the Sia modification of specific SP

a

Detection	α-2,3 linkage of sialic acid (MAA)						α-2,6 and α-2,3 linkage of sialic acid (SNA)					
Method	ELLSA	Western Blot Analysis					ELLSA	Western Blot Analysis				
Protein Sia / Apo sperm	Total	122 kDa	96 kDa	55 kDa	37 kDa	25 kDa	Total	142 kDa	65 kDa	47 kDa	40 kDa	23 kDa
Early	-0.115	-0.117	0.100	0.400	-0.517	-0.383	0.517	-0.033	-0.383	-0.383	-0.583	-0.167
Late	* 0.709	0.183	0.067	0.267	0.683	0.150	-0.103	0.300	0.550	0.683	0.667	* 0.717
Early	0.079	0.079	0.273	* 0.673	0.588	-0.273	0.115	* 0.697	0.030	* 0.733	*** 0.794	0.455
Late	-0.285	-0.370	-0.212	-0.079	-0.127	0.176	-0.127	-0.394	-0.115	-0.067	-0.079	-0.346
Early	0.567	0.433	0.383	0.200	0.143	0.200	0.100	0.033	-0.250	0.117	0.317	0.333
Late	0.133	0.433	0.383	0.200	0.143	0.200	-0.083	0.033	-0.250	0.117	0.317	0.333

b



NS	AS	OAT	NS	AS	OAT	NS	AS	OAT	NS	AS	OAT	NS	AS	OAT
TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL	TRFL
PIP	PIP	PIP	PIP	-	PIP	PIP	PIP	PIP	PIP	PIP	PIP	PIP	PIP	PIP
-	RAF1	-	RAF1	RAF1	RAF1	RAF1	-	RAF1	RAF1	RAF1	RAF1	RAF1	RAF1	RAF1
S10A9	S10A9	S10A9	S10A9	-	S10A9	S10A9	S10A9	-	S10A9	-	S10A9	S10A9	S10A9	S10A9
CLUS	CLUS	CLUS	CLUS	CLUS	CLUS	CLUS	CLUS	-	CLUS	-	-	CLUS	-	-
-	PRDX2	-	-	-	PRDX2	-	-	PRDX2	PRDX2	-	PRDX2	-	PRDX2	-
UBB	UBB	UBB	UBB	-	UBB	-	UBB	-	UBB	-	UBB	-	UBB	-
ALBU	ALBU	ALBU	ALBU	ALBU	ALBU	ALBU	ALBU	-	-	-	ALBU	-	-	ALBU
-	-	S10A8	-	-	-	S10A8	S10A8	S10A8	S10A8	-	S10A8	S10A8	-	-
-	G3P	G3P	G3P	-	-	G3P	G3P	G3P	G3P	-	G3P	-	-	-
-	CATD	-	-	-	-	-	CATD	CATD	-	-	-	CATD	-	-
-	-	-	PCDGL	-	-	PCDGL	-	-	-	-	-	-	-	-
-	-	-	PCDGM	-	-	PCDGM	-	-	-	-	-	PCDGM	-	-
-	-	-	PCDGK	-	-	PCDGK	-	-	-	-	-	PCDGK	-	-
-	-	-	-	-	-	-	-	-	-	TIMP1	-	TIMP1	TIMP1	TIMP1
CASPE	-	-	-	-	-	-	-	-	-	-	-	-	CASPE	-
CO4A	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4B	-	-	-	-	-	-	-	-	-	-	-	-	-	-
-	-	TRRAP	-	-	-	-	-	-	-	-	-	-	-	-
-	-	LEG7	-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	ROBO1	-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	APOH	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-	CD38	CD38	-	-	-	-
-	-	-	-	-	-	-	-	-	DLG5	-	-	-	-	-
-	-	-	-	-	-	-	-	-	-	-	ACTB	-	-	-
-	-	-	-	-	-	-	-	-	-	-	ACTC	-	-	-
-	-	-	-	-	-	-	-	-	-	-	-	PAEP	PAEP	-
-	-	-	-	-	-	-	-	-	-	-	-	NGAL	-	-
-	-	-	-	-	-	-	-	-	-	-	-	PEBP4	-	-
-	-	-	-	-	-	-	-	-	-	-	-	TIMP3	TIMP3	-
-	-	-	-	-	-	-	-	-	-	-	-	-	CAN2	-
-	-	-	-	-	-	-	-	-	-	-	-	-	TCPA	-
-	-	-	-	-	-	-	-	-	-	-	-	-	BNIP5	-
-	-	-	-	-	-	-	-	-	-	-	-	-	SULF1	-

(caption on next page)

Fig. 7. Correlation between sialylation of SP proteins and sperm apoptosis detected in ejaculate, and identification of proteins associated with apoptotic process. (a) The r value from correlation test is determined for each group separately (blue – NS category, orange – AS category, green – OAT category). Significant results are highlighted with a black box and the strength of significance is shown with an asterisk (*$p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). (b) The bar plots represent number (No) of proteins in normal (NS) and pathological (AS, OAT) ejaculate samples of proteins bands in different molecular weights. TRFL – Lactotransferrin, PIP – Prolactin-inducible protein, RAF1 – RAF proto-oncogene serine/threonine-protein kinase, S10A9 – Protein S100-A9, CLUS – Clusterin, PRDX2 – Peroxiredoxin-2, UBB – Polyubiquitin-B, ALBU – Albumin, S10A8 – Protein S100-A8, G3P – Glyceraldehyde-3-phosphate dehydrogenase, CATD – Cathepsin D, PCDGL – Protocadherin gamma-C4, PCDGM – Protocadherin gamma-C5, PCDGK – Protocadherin gamma-C3, TIMP1 –Metalloproteinase inhibitor 1, CASPE – Caspase 14, CO4A – Complement C4-A, CO4B – Complement C4-B, TRRAP – Transformation/transcription domain-associated protein, LEG7 – Galectin-7, ROBO1 – Roundabout homolog 1, APOH – Beta-2-glycoprotein 1, CD38 – ADP-ribosyl cyclase/cyclic ADP-ribose hydrolase 1, DLGS – Disks large homolog 5, ACTB – Actin, cytoplasmic 1, ACTC – Actin, alpha cardiac muscle 1, PAEP – Glycodelin, NGAL – Neutrophil gelatinase-associated lipocalin, PEBP4 – Phosphatidylethanolamine-binding protein 4, TIMP3 – Metalloproteinase inhibitor 3, CAN2 – Calpain-2 catalytic subunit, TCPA – T-complex protein 1 subunit alpha, BNIP5 – Protein BNIP5, SULF1 – Extracellular sulfatase Sulf-1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

proteins might be more appropriate in the future. The limit of the current approach is the accessibility of the samples and their high consumption for individual methods, especially for the ELLSA due to a small volume of ejaculate. Therefore, all results, including data from MS, are related to the spermiogram category rather than to the individual patient and for this reason we could not compare the results and correlations between individual methods of evaluation. Another limitation is the heterogeneity of glycoproteins, which makes the study of glycoproteins more challenging and complicated [2].

Mass spectrometry identified SNA detected sialoproteins that could be soluble in SP, bound to extracellular vesicles or released from spermatozoa. Protein bands marked as sialoproteins contained not only different but also the same proteins in different molecular weights, which is probably due to the protease activity during ejaculate liquefaction [20]. Not all identify proteins were sialylated. Some of these proteins can only interact with another protein in SP, such as lactotransferrin (TRFL), which acts as universal molecular transporter [21]. Additionally, the samples could be partially contaminated during collection and handling, which relates to detected proteins such as dermicidin or keratins, which are expressed by the skin but can also occur in reproductive tissue [22]. Keratin K1C16 was detected as a part of the sperm head proteome [23], which could be released from the tissue or the damaged spermatozoa into SP. According to the GlyGen database K1C16 is glycosylated by O-GlcNAcylation [14], but the sialylation of this protein has not been described yet. However, it still needs to be determined whether sialylation and the presence of proteins such as keratin K1C1 are found in the SP of normal or pathological ejaculates.

The responsibility of Sia for modulating the immune response has been outlined in both sperm and SP proteins [4,13]. The male reproductive system protects spermatozoa against the immune response and Sia could play a role in immunomodulation via its interaction with cell receptors [24,25]. The female genital tract contains Siglec receptors that recognize Sia and subsequently modulate the immune response towards spermatozoa [7]. Therefore, changes in protein sialylation may result in blocking this interaction, the suppression of immune modulation, and in consequence a formation of ASA. Total detection of sialylated glycoproteins showed that the increase of α-2,3 linked Sia correlated with the presence of ASA in ejaculate in OAT samples. Deeper examination of glycoprotein bands has shown a relationship between protein sialylation (55, 47, 40, and 23 kDa) in NS samples and the presence of both IgA and IgG ASA in the ejaculate. In these bands, a higher number of sialylated proteins was identified in NS group than in the pathological groups, which did not coincide with lectin signal intensity. These findings point towards the importance of the amount of protein sialylation rather than the number of sialylated proteins within the male reproductive tract, which contributes to the production and modification of sialylated sperm proteins during sperm development, maturation, and also ejaculation when sperm come to contact with SP [26]. Disorders that result in protein sialylation within reproductive tissues could theoretically be manifested by modification in the sialylation of SP proteins and the presence of ASA in the ejaculate. Moreover, an increased number of sialoproteins in SP could mean their removal from the surface of spermatozoa, which may thus be more susceptible to an immune reaction in the female reproductive tract.

All sialoprotein bands recognized by SNA lectin contained proteins playing a role in the immune response. One of the identified proteins is PIP, which is predicted to interact with other proteins in SP and is likely able to create multimeric protein formations [27]. This could explain the detection of this protein in several Mw. Additionally, the glycosylation could be behind its Mw increase. These assumptions are supported by the study of [27], which describes different isoelectric variants of PIP that can interact with IgG in SP. This could further indicate PIP role in immunomodulation and interaction with ASA to protect sperm against ASA binding [27]. Despite all this, we did not detect changes in sialylation of PIP between the normal and pathological samples during performed assays. However, interestingly we found SEMG1 protein and its fragments sialylated in three Mw, 40 and 23 kDa in all the normal and pathological samples with the exception of 47 kDa protein, which was detected only in NS sperm. These results are in correlation with the study of Tomar et al. [28] showing the interaction between PIP and SEMG1 in SP. In addition to PIP, SEMG1 is also able to interact with CLU based on the STRING database which we identified among the sialylated proteins and included it in the group of proteins playing a role in the immune response [29]. CLU is a highly glycosylated protein [30], which we detected in NS samples. Interestingly, AS and OAT samples were positive for CLU glycosylation only with higher Mw. This could be important in the long run for diagnostics, but a larger number of samples needs to be tested. Similarly to CLU, TRFL are proteins which could have an immunomodulatory function in the reproductive processes, and their immunomodulatory effect probably lies in their glycosylation form [13,30,31]. It is reassuring that we also detected TRFL sialylation in SP, which supports this hypothesis.

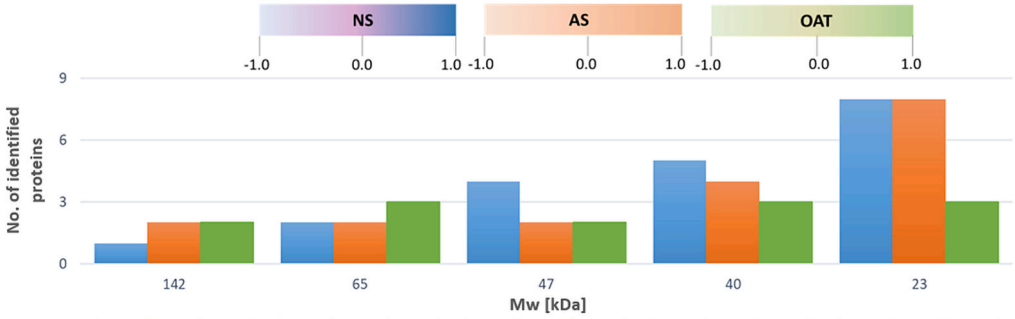
As stated earlier, the changes in sialylation of glycoproteins could be an indicator of ongoing disease or infection. Relevant to this topic, the occurrence of *E. coli* in the epididymis may cause the cleavage of Sia from spermatozoa using endogenous sialidase [9]. The presence of bacterial infection such as *Chlamydia trachomatis* could lead to ASA production or in the presence of both bacteria to sperm apoptosis [32–34]. This correlated with sialylation of SP proteins in our assays. Specifically, the higher sialylation of SP proteins, positively correlated with early apoptotic in the AS group and late apoptotic spermatozoa in the NS samples using SNA lectin detection. In addition, the lower protein sialylation correlated with the higher level of viable spermatozoa in 55 kDa protein in NS group, using MAA lectin detection. These results overall suggest that ongoing sperm apoptosis affects the presence of sialylated SP proteins. The question remains to be answered whether the changes in the sialylation of SP proteins trigger the apoptosis, or they are a consequence of apoptosis. Under physiological conditions, SP should contain antioxidants that are responsible for sperm protection against oxidative stress [35] and we detected some antioxidants in sialylated samples, such as PRDX2 which has been reported to be present on spermatozoa and in human SP [36]. Additionally, changes in sialylation together with the presence of apoptotic spermatozoa may also be a sign of an ongoing infection in the male reproductive system.

The number of identified proteins for pathological samples in apoptosis is similar or lower than in the NS samples. One of the

a

Detection	α-2,3 linkage of sialic acid (MAA)						α-2,6 and α-2,3 linkage of sialic acid (SNA)					
Method	ELLSA	Western Blot Analysis					ELLSA	Western Blot Analysis				
Protein Sia / Spermiogram	Total	122 kDa	96 kDa	55 kDa	37 kDa	25 kDa	Total	142 kDa	65 kDa	47 kDa	40 kDa	23 kDa
Volume of ejaculate	0.38	-0.077	-0.579	-0.247	-0.017	-0.434	-0.025	0.068	0.545	* 0.732	*** 0.936	0.162
Sperm count	0.249	0.150	0.067	0.367	0.667	0.133	-0.152	0.433	0.517	0.550	0.533	0.667
Sperm concentration	0.165	0.183	0.150	0.367	0.600	0.167	-0.018	0.567	0.433	0.233	0.200	0.517
Motility (WHO A+B)	0.250	0.350	-0.367	-0.050	-0.100	-0.233	0.559	0.067	0.200	0.267	0.200	-0.233
Morpho. abnormalities	-0.375	0.291	-0.385	* -0.718	-0.428	-0.068	-0.229	-0.393	-0.419	-0.333	-0.368	* -0.761
Sperm viability	-0.333	0.083	-0.383	* -0.750	-0.067	0.117	-0.261	0.250	0.100	-0.417	-0.067	-0.633
Volume of ejaculate	-0.178	-0.392	0.086	0.177	0.281	0.018	0.031	0.049	0.275	0.202	0.336	0.251
Sperm count	0.018	-0.382	-0.030	-0.042	0.115	-0.224	-0.127	0.018	0.321	-0.079	0.176	0.358
Sperm concentration	0.006	-0.340	-0.024	-0.274	-0.061	-0.176	-0.164	-0.109	0.316	-0.170	0.049	0.286
Motility (WHO A+B)	-0.128	-0.382	-0.152	-0.624	-0.552	-0.139	0.012	-0.467	0.382	-0.370	-0.115	0.333
Morpho. abnormalities	-0.451	-0.227	-0.069	0.296	0.252	0.170	-0.140	-0.031	-0.233	0.308	-0.057	-0.535
Sperm viability	0.249	0.212	-0.018	-0.224	-0.200	-0.091	0.152	0.079	0.224	-0.139	-0.273	0.188
Volume of ejaculate	-0.355	0.213	0.179	0.179	0.096	0.162	0.080	0.136	0.000	0.545	0.409	0.179
Sperm count	-0.469	* -0.800	* -0.783	-0.433	* -0.786	* -0.700	-0.176	0.400	0.467	0.467	0.333	0.333
Sperm concentration	-0.285	** -0.883	** -0.833	-0.650	* -0.786	* -0.767	-0.310	0.300	0.417	0.017	0.000	0.217
Motility (WHO A+B)	0.536	-0.092	-0.193	-0.159	-0.347	-0.126	0.519	0.134	-0.318	0.050	0.050	-0.142
Morpho. abnormalities	0.274	0.173	0.173	0.346	0.218	0.087	0.548	-0.606	-0.346	-0.346	-0.260	-0.260
Sperm viability	-0.250	-0.433	-0.383	-0.200	-0.143	-0.200	-0.017	-0.033	0.250	-0.117	-0.317	-0.333

b



NS	AS	OAT	NS	AS	OAT	NS	AS	OAT	NS	AS	OAT	NS	AS	OAT
K1C9	K1C9	K1C9	K1C9	K1C9	K1C9	K1C9	K1C9	K1C9	K1C9	K1C9	K1C9	K1C9	K1C9	K1C9
-	-	-	IPSP	IPSP	IPSP	IPSP	IPSP	IPSP	IPSP	-	-	IPSP	-	-
-	-	-	-	-	SEMG2	SEMG2	-	-	SEMG2	-	-	SEMG2	SEMG2	SEMG2
-	-	-	-	-	-	SEMG1	-	-	SEMG1	SEMG1	SEMG1	SEMG1	SEMG1	SEMG1
-	K2C8	-	-	-	-	-	-	-	-	K2C8	-	-	-	-
-	-	-	-	-	-	-	-	-	ESPB1	-	-	ESPB1	ESPB1	-
-	-	GOGA3	-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-	-	ACRO	-	-	-	-
-	-	-	-	-	-	-	-	-	-	-	H33	-	-	-
-	-	-	-	-	-	-	-	-	-	-	-	PAEP	PAEP	-
-	-	-	-	-	-	-	-	-	-	-	-	CRIS1	-	-
-	-	-	-	-	-	-	-	-	-	-	-	IMP2L	-	-
-	-	-	-	-	-	-	-	-	-	-	-	-	GGT1	-
-	-	-	-	-	-	-	-	-	-	-	-	-	TCPA	-
-	-	-	-	-	-	-	-	-	-	-	-	-	BNIP5	-
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

(caption on next page)

Fig. 8. Correlation between sialylation of SP proteins and ejaculate parameters, and identification of proteins associated with spermatozoa. (a) The r value from correlation test is determined for each group separately (blue – NS category, orange – AS category, green – OAT category). Significant results are highlighted with a black box and the strength of significance is shown with an asterisk (*$p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). (b) The bar plots represent number (No) of proteins in normal (NS) and pathological (AS, OAT) ejaculate samples of proteins bands in different molecular weights. K1C9 – Keratin, type I cytoskeletal 9, IPSP – Plasma serine protease inhibitor, SEMG2 – Semenogelin 2, SEMG1 – Semenogelin 1, K2CO – Keratin, type II cytoskeletal 8, ESPB1 – epididymal sperm-binding protein 1, ACRO – acrosin, H33 – Histone H3.3, PAEP – Glycodelin, CRIS1 – cysteine-rich secretory protein 1, IMP2L – mitochondrial inner membrane protease subunit 2, GGT1 – glutathione hydrolase 1 proenzyme, TCPA – T-complex protein 1 subunit alpha, BNIP5 – Protein BNIP5. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
Selected sialylated proteins identified by MS. Based on GlyGen database, proteins were determined their glycosylation, sialylation and the disease in which they participate. A function was assigned to individual proteins based on the Gene ontology database. Their possible role in the process of male reproduction was designed according to published studies.

Protein name/*Gene name*	Mw [kDa]	GlyGen database			GeneOntology database			Function in reproduction (based on references)
		Glyco	Sia	Disease	IR	Apo	Rep	
TRFL/*LTF*	78.182	✓	✓	Multiple cancer	✓	✓	x	Binding of molecules from SP [14] Increased expression in infertile men [15,16] Interaction with polysialic acid and possible modulation of neutrophil extracellular traps [17]
PIP/*PIP*	16.572	✓	✓	Multiple cancer	✓	✓	x	Interaction with IgG, possible immunomodulation [16,18]
CLUS/*CLU*	52.495	✓	✓	Multiple cancer	✓	✓	x	Chaperone activity, immunomodulation [19]
G3P/GAPDH	36.053	✓	✓	Multiple cancer	✓	✓	x	–
PRDX2/*PRDX2*	21.892	✓	x	Multiple cancer	✓	✓	x	–
SEMG1/*SEMG1*	52.131	x	x	Multiple cancer	✓	x	✓	Antibacterial activity [20] Modulation of motility [21] Coagulation and sperm capacitation [22]
SEMG2/*SEMG2*	65.444	✓	✓	Multiple cancer	✓	x	✓	Coagulation [23] Antibacterial activity [24] Sperm motility and capacitation [22]
CD177/*CD177*	46.363	✓	x	Multiple cancer	✓	x	x	Decreased in SP of men with oligoasthenozoospermia [25]
PAEP/*PAEP*	20.624	✓	x	Multiple cancer	x	✓	✓	Uncapacitation factor [26]
K1C9/*KRT9*	62.064	✓	✓	Multiple cancer	x	x	✓	Spermatogenesis [22]
ACRO/*ACR*	45.847	✓	x	Multiple cancer	x	x	✓	Acrosome reaction, penetration, and interaction with zona pellucida [22]
ESPB1/ELSPBP1	26.106	✓	x	Multiple cancer	x	x	✓	Sperm capacitation, fertilization [22]

TRFL/LTF – Lactotransferrin, PIP-Prolactin-inducible protein, CLUS/CLU – Clusterin, G3P/GAPDH – Glyceraldehyde-3-phosphate dehydrogenase, PRDX2 – Peroxiredoxin-2, SEMG1 – Semenogelin 1, SEMG2 – Semenogelin 2, CD177 – CD177 antigen, PAEP – Glycodelin, K1C9/KRT9 – Keratin, type I cytoskeletal 9, ACRO/ACR – Acrosin, ESPB1/ELSPBP1 – Epididymal sperm-binding protein 1; Glyco – glycosylation, Sia – Sialylation, IR – Immune Response, Apo – Apoptosis, Rep – Reproduction.

identified proteins was cathepsin D, which was detected in AS (142 and 47 kDa) and NS (23 kDa) samples, while all three mentioned proteins were correlated with sperm apoptosis. As in AS, this protein was presented in the OAT 47 kDa protein band. Cathepsin D is protein, which has been described in testes, epididymis, and on spermatozoa [37]. Further, Huo et al., [38] reported the existence of cathepsin D in more than one form which may differ by post-translational modifications. This could be the reason why we identified cathepsin D at higher Mw in pathological samples. Moreover, cathepsin D could be released from spermatozoa during the apoptosis, and correspondingly, we also detected it in SP.

Except for ASA and apoptosis, we detected an association between sialylation of SP proteins and the quality of ejaculate. SNA detected 47 and 40 kDa protein bands which were in positive correlation with the volume of ejaculate. Sia bound via α-2,3 (MAA detection) was related to the sperm count and concentration. This correlation showed that the lower level of linked Sia is associated with a higher level of sperm count in the ejaculate. The role of Sia in sperm motility has been described in connection with sperm β-defensins, which improve sperm motility [39]. Therefore, we are interested in whether the changes in SP sialylation have any relation to the sperm motility. We did not discover any correlation between the Sia detected by SNA or MAA and motility. Our results contrasted with the positive correlation of SNA detection with the sperm counts or motility reported in the study by Kratz et al. [19]. The quality of the ejaculate is also determined by sperm morphological abnormalities, which increase with a decrease in glycoprotein sialylation in SP. Since SP is a secretion of the glands of the male reproductive tract [40], we assume that disorders of the reproductive tract can also manifest themselves in proteosynthesis and post-translational modifications of SP or sperm proteins. For example, increased activity of acid glycosidases has been detected in patients with varicocele and a pathological spermiogram [41]. Conversely, varicocelectomy results in an improvement of sperm fitness and changes the expression of several proteins [42]. Prostatitis as an inflammation disorder of the prostate gland that also leads to alterations in the expression of proteins present in SP [43] and many of the proteins are also related to cancer [14]. Not only diseases, but also lifestyle choices such as smoking or obesity can affect sperm fitness and also differences in the expression of proteins into SP [44,45].

We identify sialoproteins that are associated with spermatozoa

according to the database of Gene Ontology. We expected that proteins such as golgin subfamily A member 3, acrosin or histone H3.3 are proteins which are expected to be released from the damaged spermatozoa to SP. Conversely, we also identified proteins which are already present in SP and after ejaculation interact with the sperm, such as glycodelin. Glycodelin S has been reported in SP and controls sperm capacitation [46]. Glycosylation of the glycodelin may affect the properties of this protein [47]. It has been described that seminal glycodelin S is not sialylated, while glycodelin A, which is present in the female reproductive tract is sialylated and its sialylation may initiate the apoptotic activity of this protein [48]. However, we identified glycodelin between the sialylated proteins in NS and AS group. Therefore, the question is whether, in the case of pathological conditions, if glycodelin S present in SP is also sialylated, and could be used as a marker for apoptosis or quality of ejaculate. In the future, all identified proteins need deeper examination focusing on utilization of sialylation to improve diagnostic methods for ejaculate quality.

5. Conclusion

Sialic acids are essential glycan components and are related to the fertilization processes. We hypothesized that modification in protein sialylation may indicate a pathological condition occurring in the male reproductive tract. In our study, we showed a close relationship between sialylation of SP proteins and the pathological state of the ejaculate. We confirmed this by showing that increased SP protein sialylation is associated with increased ASA in the ejaculate, increased sperm apoptosis, decreased sperm viability, and decreased sperm concentration. Additionally, a reduced concentration of sialylated SP proteins is associated with better sperm morphology. Our results indicate that when sperm are damaged and their surface proteins/glycoproteins are disrupted, whether due to diseases, such as cancer, or the presence of an infection in the male reproductive tract when the immune privilege can be disturbed and the discovery of antigens for the immune system with the ASA production and apoptotic sperm presence. Sperm damage associated with increased sialoproteins in SP is also indicated by a reduced sperm concentration in ejaculate. In contrast to sperm morphology, a sperm protein profile focusing on changes in sialylated proteins may provide deeper insight into the pathological state of the ejaculate and could be a useful diagnostic marker in clinical practice to highlight a problem in the male reproductive tract. In clinical practice, current diagnostic approaches often do not indicate pathological phenomena in the ejaculate and result in false negatives. For that reason, we proposed that the detection of SP protein sialylation could be an elegant criterion to determine ejaculate quality and for indicating pathological conditions in the male reproductive tract, which consequently could improve the treatment of male reproductive problems. Mass spectrometry identified SP sialoproteins such as semenogelins, glycodelin, prolactin-inducible protein, lactotransferrin, and clusterin, which are associated with spermatozoa and contribute to the modulation of immune response and sperm apoptosis. According to the gene ontology database, many of these proteins do not have an assigned function during reproduction. A further comprehensive study is needed to provide additional insights into the role of these sialoproteins as macromolecules contributing to successful fertilization or participating in pathological processes in reproduction.

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CRediT authorship contribution statement

Veronika Palenikova: Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Hana Pavlova:** Validation, Investigation. **Daniel Kraus:** Formal analysis, Data curation. **Zuzana Kratka:** Writing – review & editing, Resources, Data curation. **Katerina Komrskova:** Writing – review & editing, Visualization, Supervision, Funding acquisition. **Pavla Postlerova:** Writing – original draft, Visualization, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Katerina Komrskova reports financial support was provided by Ministry of Health of the Czech Republic. Veronika Palenikova reports financial support was provided by Grant Agency of the Charles University. Pavla Postlerova reports financial support was provided by Grant Agency of the Czech Republic. Veronika Palenikova, Hana Pavlova, Daniel Kraus, Katerina Komrskova, Pavla Postlerova reports financial support was provided by Institute of Biotechnology of the Czech Academy of Sciences. Veronika Palenikova, Hana Pavlova, Daniel Kraus, Katerina Komrskova, Pavla Postlerova reports financial support was provided by ERDF of BIOCEV project. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

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